Wilka Torrico De Carvalho, *Reinforcement Learning Researcher and Cognitive Scientist*

Contact Information

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Github: github.com/wcarvalho
Google Scholar

Research Interests

My research aims to build AI that helps us live more rewarding lives. To get there, I study deep reinforcement learning in naturalistic settings—enabling us to both produce capable agents and to characterize the human reinforcement learning algorithm. My research spans continual learning, long-horizon RL, multi-agent collaboration with humans, and open-source infrastructure for human-AI experiments. My long-term goal is to develop AI that promotes human flourishing.

Employment

Harvard, Boston, MA — *Sept 2023 –*
Kempner Fellow

Google DeepMind, London, UK — *Sept 2022 – May 2023*
Research Scientist Intern

Google DeepMind, London, UK — *Mar 2021 – July 2021*
Research Scientist Intern

Microsoft Research, Redmond, WA — *June 2018 – Aug 2018*
Research Scientist Intern

IBM Research, San Jose, CA — *Sept 2017 – Dec 2017*
Research Scientist Intern

VISA Research, Palo Alto, CA — *May 2017 – Aug 2017*
Research Scientist Intern

Education

University of Michigan–Ann Arbor, Ann Arbor, MI — Sep 2018 - July 2023
School of Engineering, Ph.D. in Computer Science
Advisors: Satinder Singh, Honglak Lee, Richard Lewis

University of Southern California, Los Angeles, CA — Aug 2015 - May 2017
Viterbi School of Engineering, M.S. in Computer Science
Advisor: Yan Liu

Stony Brook University, Stony Brook, NY — Aug 2011 - May 2015
College of Arts and Sciences, B.S. in Physics
Advisor: Axel Drees

Brooklyn Technical High School, Brooklyn, NY — May 2007 - May 2011
Diploma in Applied Physics

Honors & Awards

GitLab Foundation AI for Economic Opportunity Award Finalist (7% selected)	2025
Kempner Institute Fundamentals of Intelligence Fellowship	2023
ICML, NeurIPS Top Reviewer	2022, 2023
University of Michigan "The Glass is Half-Full" Award	2019
Hertz Fellowship Finalist (10% selected)	2019
1/200 chosen internationally for Heidelberg Laureate Forum	2018
GEM National Fellowship sponsored by IBM, Adobe	2017, 2018
University of Michigan Rackham Merit Fellowship	2017

ICLR, Neurips BAI Travel Award	2017, 2019
NSF Graduate Research Fellowship (12% selected)	2015
Provost Award for Academic Excellence (20/5000 graduates chosen)	2015
Researcher of the Month (1 monthly at Stony Brook University)	2014
HHMI Minority Undergraduate Research Fellowship	2014
ΣΠΣ Physics Honor Society (sponsored by Alfred Goldhaber)	2013
NSF Louis Stokes Alliance for Minority Participation Scholar	2011

Preprints

Purab Seth*, Neil Shah*, Kunal Jha, Samuel J. Gershman, Max Kleiman-Weiner, **Wilka Carvalho**. *"Task diversity produces systematic transfer but inhibits continual reinforcement learning."* Under review at RLC. 2026

Wilka Carvalho, Sam Hall-McMaster, Honglak Lee, Samuel J. Gershman. *"Preemptive Solving of Future Problems: Multitask Preplay in Humans and Machines."* Under review at the Proceedings of the National Academy of Sciences (PNAS). 2026

Wilka Carvalho*, Andrew Lampinen*. *"Naturalistic Computational Cognitive Science: Towards Generalizable Models and Theories That Capture The Full Range Of Natural Behavior."* Under review at Psychological Review. 2026

Publications

Kunal Jha, **Wilka Carvalho**, Yancheng Liang, Simon S Du, Max Kleiman-Weiner, Natasha Jaques. *"Cross-environment Cooperation Enables Zero-shot Multi-agent Coordination."* In **ICML**, 2025 **(Oral)**

Wilka Carvalho*, Momchil S Tomov*, William de Cothi*, Caswell Barry, Samuel J. Gershman. *"Predictive representations: building blocks of intelligence."* In **Neural Computation**, 2024

Wilka Carvalho, Andre Saraiva, Angelos Filos, Andrew Lampinen, Loic Matthey, Richard L Lewis, Honglak Lee, Satinder Singh, Danilo Jimenez Rezende, Daniel Zoran. *"Combining behaviors with the successor features keyboard."* In **NeurIPS**, 2024

Wilka Carvalho, Andrew Lampinen, Kyriacos Nikiforou, Felix Hill, Murray Shanahan. *"Feature-Attending Recurrent Modules for Generalization in Reinforcement Learning."* In **TMLR**, 2023

Wilka Carvalho, Angelos Filos, Richard L. Lewis, Honglak Lee, Satinder Singh. *"Composing Task Knowledge with Modular Successor Feature Approximators."* In **ICLR**, 2023

Lajanugen Logeswaran, **Wilka Carvalho**, Honglak Lee. *"Learning Compositional Tasks from Language Instructions."* In **AAAI**, 2023

Chris Hoang, Sungryull Sohn, Jongwook Choi, **Wilka Carvalho**, Honglak Lee. *"Successor Feature Landmarks for Long-Horizon Goal-Conditioned Reinforcement Learning."* In **NeurIPS**, 2021

Wilka Carvalho, Anthony Liang, Kimin Lee, Sungryull Sohn, Honglak Lee, Richard L. Lewis, Satinder Singh. *"Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in First-person Simulated 3D Environments."* In **IJCAI**, 2021

Wilka Carvalho*, Sanjay Purushotham*, Tanachat Nilanon, Yan Liu. *"Variational Recurrent Adversarial Domain Adaptation."* In **ICLR**, 2017

Workshop Publications

Wilka Carvalho, Vikram Goddla, Ishaan Sinha, Hoon Shin, Kunal Jha. *"NiceWebRL: a Python library for human subject experiments with reinforcement learning environments."* In Human-AI Complementarity for Decision Making Workshop, 2025 **(Oral)**

Wilka Carvalho, Vikram Goddla, Ishaan Sinha, Hoon Shin, Kunal Jha. *"NiceWebRL: a Python library for human subject experiments with reinforcement learning environments."* In New York RL Workshop, 2025 **(Oral)**

Purab Seth*, Neil Shah*, **Wilka Carvalho**. *"Learning Compositional Tasks from Language Instructions."* In Reinforcement Learning and Decision Making Workshop, 2025

Lajanugen Logeswaran, **Wilka Carvalho**, Honglak Lee. *"Learning Compositional Tasks from Language Instructions."* In NeurIPS Deep RL Workshop, 2021

Wilka Carvalho, Murray Shanahan. *"Learning to Represent State with Perceptual Schemata."* In ICML {Unsupervised RL, Real Life RL} Workshops, 2021

Wilka Carvalho, Anthony Liang, Kimin Lee, Sungryull Sohn, Honglak Lee, Richard L. Lewis, Satinder Singh. *"Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in First-person Simulated 3D Environments."* In {ICML Object-Oriented Learning, NeurIPS DeepRL} Workshops, 2020

Chris Hoang, Sungryull Sohn, Jongwook Choi, **Wilka Carvalho**, Honglak Lee. *"Successor Landmarks for Efficient Exploration and Long-Horizon Navigation."* In NeurIPS DeepRL Workshop, 2020

Wilka Carvalho, Kimin Lee, Anthony Liang, Ryan Krueger, Richard L. Lewis, Satinder Singh, Honglak Lee. *"Efficiently Learning to Perform Household Task with Object-oriented Exploration."* In NeurIPS BAI, 2019 **(Oral)**

Bryant Chen*, **Wilka Carvalho***, Benjamin Edwards, Taesung Lee, Ian Molloy, Heiko Ludwig. *"Detecting Backdoor Attacks on Deep Neural Networks by Activation Clustering."* In AAAI Artificial Intelligence Safety Workshop, 2018 **(Best Paper)**

Sanjay Purushotham*, **Wilka Carvalho***, Yan Liu. *"Variational Adversarial Deep Domain Adaptation for Health Care Time Series Analysis."* In NeurIPS Machine Learning for Healthcare Workshop, 2016 **(Spotlight)**

Wilka Carvalho. *"Modeling a Detection of internally reflected Cherenkov light (DIRC) Particle Detector for High-Multiplicity Collisions."* In SURC, 2015

Patents

Wilka Carvalho, Angelos Filos, Danilo Jimenez Rezende, Daniel Zoran. *"Controlling Agents by Transferring Successor Features To New tasks."* 2023

Wilka Carvalho. *"Compositional Generalization in Reinforcement Learning."* 2023

Bryant Chen, **Wilka Carvalho**, Heiko Ludig, Ian Molloy, Jialong Zhang, Benjamin Edwards. *"Detecting poisoning attacks on neural networks by activation clustering."* 2020

Wilka Carvalho, Bryant Chen, Benjamin Edwards, Taesung Lee, Ian Molloy, Jialong Zhang. *"Using Gradients to Detect Backdoors in Neural Networks."* 2018

Wilka Carvalho, Yan Liu, Tanachat Nilanon, Sanjay Purushotham. *"Effective Knowledge Transfer Among Patient Populations via Deep Learning."* 2017

Invited Talks

NiceWebRL: A Python Library for Human Experiments with Reinforcement Learning Environments

Human-AI Complementarity for Decision Making Workshop. *Carnegie Mellon University.* (September, 2025)
New York Reinforcement Learning Workshop. *NYC.* (September, 2025)
Cognitive Computational Neuroscience Conference. *Amsterdam.* (August, 2025)

Naturalistic Computational Cognitive Science
Cognitive Computational Neuroscience Conference. *Amsterdam.* (August, 2025)

Scaling Up Reinforcement Learning Theories of Cognition
Cognitve Science Colloquiam Series. *University of Maryland at College Park.* (December, 2026)
Machine Intelligence through Decision-making and Interaction Lab. *UT Austin.* (September, 2025)
Cognition, Brain, and Behavior Seminar. *Harvard University.* (April, 2025)
Center for Theoretical Neuroscience. *Columbia University.* (April, 2025)
Mattar Lab. *NYU.* (April, 2025)
Emerge Lab. *NYU.* (April, 2025)
Computation and Decision-Making Lab. *NYU.* (April, 2025)
RL Seminar. *Tübingen.* (March, 2025)
Google DeepMind. *San Francisco.* (March, 2025)
Cognition Seminar. *UC Berkeley.* (March, 2025)
The Rational Altruism Lab. *UCLA.* (March, 2025)
Social and Decision Neuroscience Seminar. *Caltech.* (March, 2025)
Symposium on "Frontiers of Machine Learning and AI. *USC.* (March, 2025)
Neurolunch Seminar. *Harvard.* (March, 2025)
RL Seminar. *UMich.* (February, 2025)
Vision Lab. *Harvard University.* (February, 2025)
Sabatini Lab. *Harvard University.* (February, 2025)

Predictive Representations: Building Blocks of Intelligence
Computational Cognitive Science Laboratory. *UC Berkeley.* (March, 2025)
Computational Cognitive Neuroscience Lab. *Harvard University.* (February, 2025)
Neuro Systems Club. *Harvard University.* (Fall, 2024)
Mathis Group for Computational Neuroscience and AI. *EPFL.* (Fall, 2024)
Theory of Learning Lab. *Gatsby Computational Neuroscience Unit (UCL).* (Summer, 2024)
Conference on the Mathematics of Neuroscience and AI. *Rome, Italy.* (Summer, 2024)
Theoretical Cognitive Neuroscience Lab. *Boston University.* (Fall, 2023)

Towards Human-like Deep Reinforcement Learning
Computational Cognitive Science Group. *Princeton.* (Fall, 2023)
Kempner Institute. *Harvard University.* (March, 2023)

Composing Task Knowledge with Modular Successor Feature Approximators
RL at Harvard. *Harvard University.* (August, 2023)

Weak Inductive Biases for Composable Primitive Representations
Computational Neuroscience Group. *UCL.* (October, 2022)
Intelligent Robot Lab. *Brown.* (February, 2022)
Computational Cognitive Neuroscience Lab. *Harvard University.* (February, 2022)
Computational Cognitive Neuroscience Lab . *University of California—Berkeley.* (January, 2022)
Deep Learning Community of Practice. *Intel.* (January, 2022)
Deep RL Workshop. *NeurIPS.* (December, 2021)

Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in a First-Person Simulated 3D Environment
Object Representations for Learning Workshop. *NeurIPS.* (December, 2020)

Cognitive Tools Lab. *University of California—San Diego.* (November, 2020)
Cognitive Science Community. *University of Michigan.* (November, 2020)
Stony Brook Society of Physics. *Stony Brook, NY.* (April, 2020)

Variational Recurrent Adversarial Deep Domain Adaptation
Machine Learning Lunch Seminar. *University of Southern California.* (April, 2017)

Teaching Experience

Harvard University, Cambridge, MA
Teaching Assistant, *Fall 2025*
Building Transformers from Scratch Workshop

Stony Brook University, Stony Brook, NY
Calculus Instructor, *Spring 2015*
Worked with two math professors to develop and teach a supplementary calculus curriculum that promoted minority representation in stem majors.

Stony Brook University, Stony Brook, NY
Educational Opportunity Program Personal Tutor, *Spring 2013 - Fall 2014*
Tutored marginalized students in introductory physics and math courses

Service

Action Editor, Transactions on Machine Learning Research, 2023 -
Area Chair, Reinforcement Learning Conference, 2023 -
Reviewer, Royal Society Open Science, 2025 -
Reviewer, Journal on Machine Learning Research, 2025 -
Reviewer, Perspectives on Psychological Science, 2025 -
Reviewer, Neural Computation, 2023 -
Reviewer, ICLR, 2021 -
Reviewer, ICML, 2021 -
Reviewer, NeurIPS, 2020 -
Student Volunteer, ICLR, 2017

Outreach

Lecture for summer program for disadvantaged students, Dorchester, MA, 2024
DAPCEP (Detroit area pre college engineering program), Michigan, 2023
Stony Brook's CSTEP Alumni mentoring program, Michigan, 2022
HS2 Engineer's Highschool Panel, Virtual, 2021
Stony Brook Society of Physics, Stony Brook, NY, 2020
Michigan Explore Graduate Studies Workshop, Ann Arbor, MI, 2019
Research and Fellowships Week NSF Panel, Los Angeles, CA, 2016
National Society of Black Engineers Grad Panel, Los Angeles, CA, 2016
Graduate School External Fellowship Boot Camp, Los Angeles, CA, 2016
Mentored marginalized high school youth through the Pullias Center for Higher Education, Los Angeles, CA, 2016
Engineering Graduate Diversity Symposium, Los Angeles, CA, 2015
Black Student Association: What it takes to go to Graduate School, Los Angeles, CA, 2015
Collegiate Science and Technology Entry Program Undergraduate Research Panel, Stony Brook, CA, 2014

Press

Finding Gaps in Your Grad School Apps
This Week In Machine Learning (TWIML) Research Feature
University of Michigan Research Feature
Exploring the source of social stereotypes
Black History Month: Why a career in science?

Research Feature by the USC Graduate School
2015 NSF Graduate Research Fellow Wilka Carvalho
Biomath Learning Center Launches Modified Supplemental Instruction Program
URECA Research of the Month: Wilka Carvalho
Student Feature by Stony Brook University

Interests • traveling • chess • software development • improvisational dance • deadpan humor